



BIOSOLIDS LAND APPLICATION SITE IDENTIFICATION FORM

*Authorized under 1994 PA 451, as amended. Completion of form is required.
Applicant may be subject to civil and/or criminal penalties for providing false information.*

Complete the following information and provide the attachments listed on the second page for each new or updated biosolids land application site. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e., new landowner).

For additional information, Biosolids Program Staff are listed at: https://www.michigan.gov/documents/deq/wrd-biosolids-staff_402800_7.pdf.

Facility Name City of Cadillac WWTP		NPDES Permit or COC Number MI0020257		Request: New Site / Site Update / Site transfer Site Update	
Site ID Name (ex T/R/S – Owner (05N08E03-LB01) 21n09w25-ab01			Site location description (street name, nearest crossroad) West of Seeley Rd, North of 50 Rd		
County Wexford			Township Clam Lake		
(ex: NW, NE, SW, SE) SE 1/4		(ex: NW, NE, SW, SE) SE 1/4		Township (ex: 05N) T21N	Range (ex: 08E) R09W
Latitude (ex: N 42.09651) 44.18164 N		Longitude (ex: -81.09651) -85.33745 W		Method Used (i.e., GPS, topo) Google Earth	
Name of Owner Alan Benson			Phone 231-920-6783		
Street or P.O. Box 11799 East 50 Mile Rd			City or Town Cadillac		State MI
			Zip 49601		
Is there a written agreement from the owner to land apply? ☒ Yes ☐ No (written agreement is required)					
Previous Landowner's Name (if site update) Hartley Benson			Property Purchase Date N/A		
Name of Lessee/Farmer (if different)			Phone		
Street or P.O. Box			City or Town		State
					Zip
Site Type ☒ Agricultural ☐ Forest ☐ *Reclamation Site (separate approval is required) ☐ Lawn/Home Garden (requires EQ Biosolids)					
Soils Information Date of last soils analysis <u>9-27-2021</u> *Phosphorous <u>78</u> units ppm Potassium <u>123</u> units ppm pH <u>7.1</u> Soil type(s) <u>Sandy</u> *300 lb/ac (or 150 ppm) Bray P1 maximum limit			Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☒ *Higher (Requires an approved site reclamation plan)		
Expected Crops/Vegetation to be grown Crop/Vegetation: <u>Corn</u> Nitrogen Requirement: <u>135</u> lbs. N/acre Crop/Vegetation: _____ Nitrogen Requirement: _____ lbs. N/acre Crop/Vegetation: _____ Nitrogen Requirement: _____ lbs. N/acre					
Total acreage of site 46.2		Acreage used for crops 46.2		Acreage used for land application	
Does this site have a high potential for public exposure? ☐ Yes ☒ No			If agricultural, is the site tilled? ☐ Yes ☐ No ☒ NA		

6100 W. Blue Rd. (M-55)
Lake City



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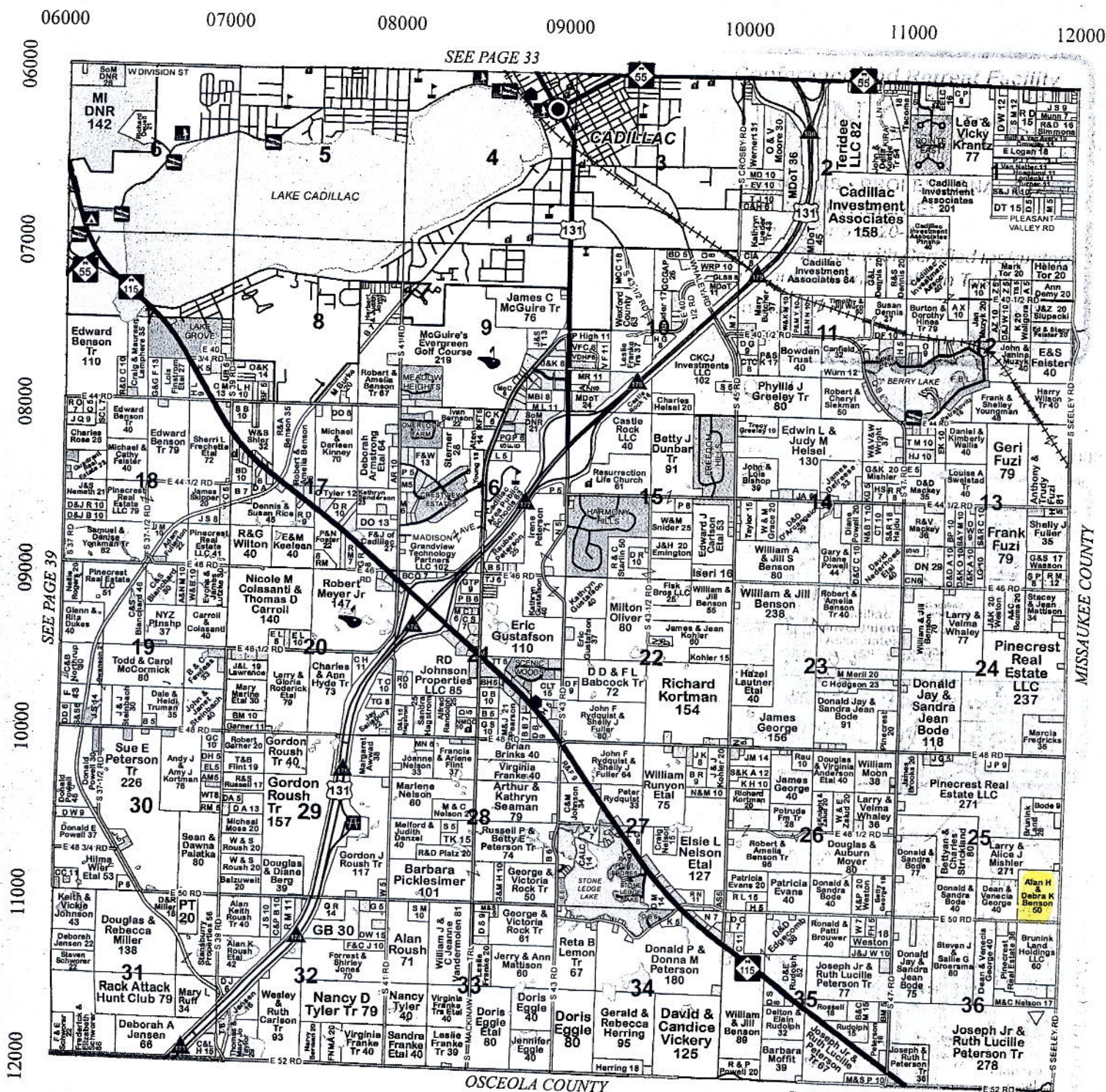
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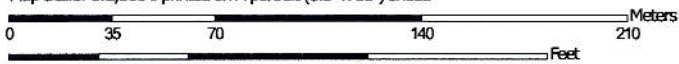
CLAM LAKE T21N • R9W



Land Application of Municipal Sewage Sludge—Lake and Wexford Counties, Michigan



Map Scale: 1:2,500 if printed on A portrait (8.5" x 11") sheet.



Map projection: Web Mercator Corner coordinates: WGS84 Edge tics: UTM Zone 16N WGS84



Natural Resources
Conservation Service

Web Soil Survey
National Cooperative Soil Survey

12/16/2024
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MAP LEGEND

Area of Interest (AOI)	Background
 Area of Interest (AOI)	 Aerial Photography
Soils	
Soil Rating Polygons	
 Very limited	
 Somewhat limited	
 Not limited	
 Not rated or not available	
Soil Rating Lines	
 Very limited	
 Somewhat limited	
 Not limited	
 Not rated or not available	
Soil Rating Points	
 Very limited	
 Somewhat limited	
 Not limited	
 Not rated or not available	
Water Features	
 Streams and Canals	
Transportation	
 Rails	
 Interstate Highways	
 US Routes	
 Major Roads	
 Local Roads	

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:15,800.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL:
Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Lake and Wexford Counties, Michigan
Survey Area Data: Version 18, Aug 28, 2024

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: May 27, 2023—May 28, 2023

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Land Application of Municipal Sewage Sludge

Map unit symbol	Map unit name	Rating	Component name (percent)	Rating reasons (numeric values)	Acres in AOI	Percent of AOI
12C	Emmet-Montcalm complex, 6 to 12 percent slopes	Somewhat limited	Emmet (50%)	Too acid (0.14)	7.2	43.4%
				Slope (0.04)		
			Montcalm (40%)	Too acid (0.42)		
				Droughty (0.07)		
				Slope (0.04)		
12D	Emmet-Montcalm complex, 12 to 18 percent slopes	Very limited	Emmet (50%)	Slope (1.00)	7.2	43.4%
				Too acid (0.14)		
			Montcalm (40%)	Slope (1.00)		
				Too acid (0.42)		
				Droughty (0.07)		
			Kalkaska (4%)	Filtering capacity (1.00)		
				Slope (1.00)		
				Droughty (0.77)		
				Too acid (0.67)		
			East Lake (3%)	Filtering capacity (1.00)		
				Slope (1.00)		
				Droughty (0.96)		
			Nester (3%)	Slow water movement (1.00)		
				Slope (1.00)		
12E	Emmet-Montcalm complex, 18 to 40 percent slopes	Very limited	Emmet (50%)	Slope (1.00)	2.2	13.2%
				Too acid (0.14)		
			Montcalm (44%)	Slope (1.00)		
				Too acid (0.42)		
				Droughty (0.07)		
			Kalkaska (3%)	Filtering capacity (1.00)		
				Slope (1.00)		
				Droughty (0.77)		
				Too acid (0.67)		
			East Lake (3%)	Filtering capacity (1.00)		
				Slope (1.00)		

Map unit symbol	Map unit name	Rating	Component name (percent)	Rating reasons (numeric values)	Acres in AOI	Percent of AOI
				Droughty (0.96)		
Totals for Area of Interest					16.7	100.0%

Rating	Acres in AOI	Percent of AOI
Very limited	9.4	56.6%
Somewhat limited	7.2	43.4%
Totals for Area of Interest	16.7	100.0%

Description

Application of sewage sludge not only disposes of waste material but also can improve crop production by increasing the supply of nutrients in the soils where the material is applied. Sewage sludge is the residual product of the treatment of municipal sewage. The solid component consists mainly of cell mass, primarily bacteria cells that developed during secondary treatment and have incorporated soluble organics into their own bodies. The sludge has small amounts of sand, silt, and other solid debris. The content of nitrogen varies. Some sludge has constituents that are toxic to plants or hazardous to the food chain, such as heavy metals and exotic organic compounds, and should be analyzed chemically prior to use.

The content of water in the sludge ranges from about 98 percent to less than 40 percent. The sludge is considered liquid if it is more than about 90 percent water, slurry if it is about 50 to 90 percent water, and solid if it is less than about 50 percent water.

The ratings are based on the soil properties that affect absorption, plant growth, microbial activity, erodibility, the rate at which the sludge is applied, and the method by which the sludge is applied. The properties that affect absorption, plant growth, and microbial activity include saturated hydraulic conductivity (Ksat), depth to a water table, ponding, the sodium adsorption ratio, depth to bedrock or a cemented pan, available water capacity, reaction, salinity, and bulk density. The wind erodibility group, soil erosion factor K, and slope are considered in estimating the likelihood that wind erosion or water erosion will transport the waste material from the application site. Stones, cobbles, a water table, ponding, and flooding can hinder the application of sludge. Permanently frozen soils are unsuitable for waste treatment.

The ratings are both verbal and numerical. Rating class terms indicate the extent to which the soils are limited by all of the soil features that affect agricultural waste management. "Not limited" indicates that the soil has features that are very favorable for the specified use. Good performance and very low maintenance can be expected. "Somewhat limited" indicates that the soil has features that are moderately favorable for the specified use. The limitations can be overcome or minimized by special planning, design, or installation. Fair performance and moderate maintenance can be expected. "Very limited" indicates that the soil has one or more features that are unfavorable for the specified use. The limitations generally cannot be overcome without major soil reclamation, special design, or expensive installation procedures. Poor performance and high maintenance can be expected.

Numerical ratings indicate the severity of individual limitations. The ratings are shown as decimal fractions ranging from 0.01 to 1.00. They indicate gradations between the point at which a soil feature has the greatest negative impact on the use (1.00) and the point at which the soil feature is not a limitation (0.00).

The map unit components listed for each map unit in the accompanying Summary by Map Unit table in Web Soil Survey or the Aggregation Report in Soil Data Viewer are determined by the aggregation method chosen. An aggregated

rating class is shown for each map unit. The components listed for each map unit are only those that have the same rating class as listed for the map unit. The percent composition of each component in a particular map unit is presented to help the user better understand the percentage of each map unit that has the rating presented.

Other components with different ratings may be present in each map unit. The ratings for all components, regardless of the map unit aggregated rating, can be viewed by generating the equivalent report from the Soil Reports tab in Web Soil Survey or from the Soil Data Mart site. Onsite investigation may be needed to validate these interpretations and to confirm the identity of the soil on a given site.

Rating Options

Aggregation Method: Dominant Condition

Component Percent Cutoff: None Specified

Tie-break Rule: Higher



CITY OF CADILLAC WASTEWATER TREATMENT PLANT

BIOSOLIDS RECYCLING PROGRAM PARTICIPATION

I hereby agree to participate in the City of Cadillac WWTP biosolids recycling program. I understand that biosolids will be applied to my land in accordance with federal EPA regulations and guidelines of the Michigan Department of Environmental Quality.

I understand that biosolids have been analyzed to be sure they can be applied to agricultural land in accordance with federal regulations and MDEQ guidelines. I understand that biosolids contain some heavy metals and that they must be below the EPA allowable limits for land application. I understand that the nutrients and/or the metal concentration may limit the frequency of biosolids application to my land.

My land is used for the production of agricultural crops. Information concerning crop and site restrictions has been explained to me. I agree to allow representatives of the MDEQ, Cadillac Waste Water Treatment Plant and the City's Injection contractor to have access to my land to obtain soil samples and for application of biosolids.

If I decide to cancel this agreement, I will give 60 days advance notice to the City of Cadillac WWTP.

Landowner: Alan Benson Signed: Alan Benson

Address: 11799 East 50th Rd Date: 10-29-19

Township: Clam Lake Section: 25 Acres: 50

City WWTP Field Number: 21N09W25-AB01

If Rented, Farmer: _____

Address: _____

To: CITY OF CADILLAC
200 N LAKE ST
CADILLAC, MI 49601-2800

For: FALL 2023 APPLICATIONS

Attn: PATRICK MILLER

SOIL TEST REPORT

Date Received: 10/17/2023

Date Reported: 10/19/2023 Page: 1 of 1

Sample ID	Lab Number	Organic Matter %	Phosphorus		Potassium K ppm	Magnesium Mg ppm	Calcium Ca ppm	Sodium Na ppm	Soil pH	Buffer pH	CEC meq/100g	Percent Cation Saturation				
			Bray-1 Equiv ppm-P	Bray P2 ppm-P								% K	% Mg	% Ca	% H	% Na
AB01WEX	9825	2.1	110 VH	147 VH	172 H	165 H	850 M		7.0		6.1	7.3	22.7	70.1		
AB02MIS	9826	2.1	169 VH	232 VH	247 VH	195 H	950 M		7.0		7.0	9.0	23.2	67.8		
AB01MIS	9827	2.3	152 VH	222 VH	186 H	175 H	950 M		6.8		6.9	6.9	21.2	68.9	3.0	
DG01WEX	9828	2.4	57 VH	69 H	100 M	185 H	900 M		6.7	6.9	7.5	3.4	20.6	60.0	16.0	
DG01MIS	9829	2.3	82 VH	99 VH	141 H	195 H	950 M		6.9		6.8	5.3	23.8	69.5	1.5	

[illegible]

Report reviewed and approved by our professional agronomy staff.

A&L-R

